

Six-Rung Substrate-Mediator Ladder Completed at Population Scope

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 14, 2025

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

The Coordination Knowledge Substrate (CKS) theory series — three papers published across Paper 1, Paper 2, and Paper 3 — constitutes a unified architectural development rather than three independent contributions. This note formalizes the structural observation that makes that unity visible: the same architectural primitive, human-governed substrate as coordination medium, operates at six progressively larger coordination scopes across the trilogy. Each scope is one rung of a substrate-mediator ladder. Rungs 1 through 5 are stated architectural commitments, directly defended by the core-theory claims of Papers 1, 2, and 3. Rung 6 is Paper 3’s extension claim (Claim 6), which extends the ladder to population-scale collective evolution in the calibrated-humility register. The ladder as a prior-art structure establishes three things: that the trilogy is one architectural development; that each rung’s governance architecture inherits from prior rungs without discontinuity; and that any implementation of coordination at any rung — including inter-Self and population-scale coordination — is implementing what the trilogy’s architectural commitment already describes. This note formalizes the ladder framework as a named prior-art structure and previews the Series T Phase T2 notes that will formalize each rung-to-rung transition individually.

1. The architectural primitive and the question of scope

The CKS theory series commits to one architectural primitive at its foundation: the human-governed substrate as coordination medium. In this commitment, the coordination knowledge of a human–AI system — its entities, relationships, decisions, conflicts, rationale, and governance rules — lives in a

persistent structured artifact outside the LLM. Humans retain the authority to inspect, modify, and override that artifact at any time. The LLM acts as a mediator of the substrate's content rather than as the owner or author of coordination state. This is the architecture Paper 1 establishes at the cell level, Paper 2 extends to the multi-cell Self, and Paper 3 extends to inter-Self and population scope.

The question the ladder formalization answers is: at how many scopes does this architectural primitive operate, and what is the relationship among those scopes? The answer is not that the primitive is replicated at each scope as a fresh invention. The answer is that one architectural commitment composes to progressively larger coordination contexts when extended systematically. Each extension inherits the prior scope's commitments without replacing them. The resulting structure — six scopes, one primitive, cumulative inheritance — is the substrate-mediator ladder.

Formalizing this structure as a named prior-art framework does work that the three papers alone do not fully accomplish. The papers establish their commitments at their respective scopes. The ladder makes explicit that the three papers are stages in a single architectural development, that no rung is a novel departure from prior rungs, and that any claim to novelty for any individual rung encounters the prior art not only of that rung's own paper but of the entire ladder beneath it.

2. The six rungs

Rung 1 — Single human ↔ LLM within one cell

Source: Paper 1, core theory (Claims 1–6, §§2–8).

Scope: The foundational coordination unit: one human (or a small group acting as a single governance principal) and one LLM operating through a governed cell. The cell contains a human-governed substrate that carries structured coordination knowledge. The LLM reads from and writes to the substrate as a mediator; it does not own or author the substrate. Human governance consists in the authority to inspect, modify, and override any substrate content or orchestration rule at any time.

Architectural content established at this rung: All six Paper 1 commitments — substrate as coordination artifact, human governance as authority rather than labor, conflict preservation as first-class substrate state, AI as substrate mediator, tool-agnosticism at the substrate layer, and linear-cost scaling of the governance architecture — are established here. These six commitments are the load-bearing inheritance that every subsequent rung carries forward.

Prior-art significance: Rung 1 is the most fully specified rung. Every higher rung's architectural properties derive from this rung's commitments. Any implementation of human-governed AI coordination — at any scope — that satisfies Rung 1's six commitments is building on what this rung already establishes.

Rung 2 — Multiple humans ↔ LLM within one cell

Source: Paper 1, extension claim (Claim 6).

Scope: Multiple humans coordinating through one governed cell with one LLM. The cell substrate remains the coordination medium; the extension is to multi-human participation within that cell.

Architectural content established at this rung: The linear-cost commitment from Rung 1 extends to multi-human participation without compounding governance costs. Adding participants to a governed cell does not require a proportional increase in governance overhead; the architecture scales by design. Paper 1 Claim 6 positions this as an extension of the Rung 1 architecture, not a new architecture.

Inheritance from Rung 1: The six Paper 1 commitments carry forward unchanged. Multi-human participation does not alter the substrate's role, the governance authority structure, or the LLM's mediator position. It scales Rung 1 horizontally within the cell boundary.

Rung 3 — Cell ↔ aspect ↔ Self within one Self

Source: Paper 2, core theory (Claims 1–6, §§3–8).

Scope: Multiple cells organized into aspects, integrated into one Self. The Self is a three-level architecture — cells at the base, aspects grouping related cells, the Self integrating aspects into a unified coordination structure. A Self is not a larger cell; it is a hierarchical organization of cells with an instinct/reasoning separation that locates stable behavioral dispositions in the LLM layer while coordination governance lives in the substrate layer.

Architectural content established at this rung: Paper 2 establishes that the substrate-mediator commitment extends to the three-level architecture without modification. Governance at the Self level operates through the same substrate-content mechanisms that govern individual cells; the authority structure is not replaced by a different governance model at larger scope. Paper 2's instinct/reasoning separation — the finding that the LLM's instinct layer is a substrate-content pattern rather than an internal LLM property — is the key architectural move that makes the substrate-mediator commitment coherent at Self scope.

Inheritance from Rungs 1–2: All Paper 1 commitments carry into Paper 2 as its architectural foundation. Paper 2's core theory extends, not replaces, what Rungs 1–2 establish.

Rung 4 — Self as enterprise architecture

Source: Paper 2, extension claim (Claim 6).

Scope: The Self scales to enterprise integration scope. Multiple organizational functions, spanning what would conventionally be described as enterprise systems, are unified under one Self architecture. Paper 2 Claim 6 explicitly frames this as an extension claim projecting the three-level architecture onto enterprise-scale coordination.

Architectural content established at this rung: Enterprise scope does not require a new governance model. The substrate-mediator commitment, with human governance at the authority layer and the LLM as mediator, applies at enterprise scope under the same architectural principles that apply at cell

scope. Enterprise integration is a scaling of Rung 3's three-level architecture, not a departure from it.

Inheritance from Rungs 1–3: The enterprise extension is structurally parallel to Rung 2's intra-cell multi-human extension: a Claim 6 extension that projects the prior rung's architecture onto a larger scope while retaining all prior commitments.

Rung 5 — Self ↔ Self via shared substrate

Source: Paper 3, core theory (Claims 1–5, §§4–8).

Scope: Multiple Selves coordinating through a shared substrate. Each Self operates its own internal substrate and LLM architecture. The shared substrate is the coordination medium at inter-Self scope — it carries the coordination knowledge that crosses organizational or deployment boundaries. Human governance at inter-Self scope operates through joint authority across the governance perimeters of the participating Selves.

Architectural content established at this rung: Paper 3 Claims 1–5 establish that the substrate-mediator commitment extends to inter-Self coordination without modification in kind. The shared substrate is a governed substrate in the CKS sense; the LLMs of the participating Selves act as mediators of the shared substrate's content; human governance consists in joint authority over the shared substrate's content and rules. Inter-Self coordination does not require an orchestration layer that bypasses human governance; it requires a shared substrate that satisfies the same architectural commitments the cell-level substrate satisfies.

Inheritance from Rungs 1–4: All Paper 1 commitments — substrate as coordination artifact, human governance as authority, conflict preservation, AI as mediator, tool-agnosticism, linear-cost composition — carry forward at inter-Self scope with within-scope rendering. Conflict preservation, for example, carries forward as first-class addressable shared-substrate state; tool-agnosticism holds at the shared substrate's structured-state-plus-access requirements rather than at one cell's.

Rung 6 — Population-scale collective evolution (calibrated-humility register)

Source: Paper 3, extension claim (Claim 6, §9).

Scope: A network of Selves accumulating full-aspect-integration events at population scope. When many CKS-governed Selves operate across many organizations, the accumulated coordination events across all of them constitute a population-level substrate of governed coordination knowledge. Paper 3 Claim 6 articulates this as the architectural possibility the preceding five claims compose to at population register.

Architectural content at this rung: The substrate-mediator commitment, operating at population scope, surfaces the possibility that three population-level dynamics emerge from accumulated inter-Self coordination: substrate-content commitment at population scope; population-scale evolution dynamics fed by accumulated events across many Selves; and capability-motivated differentiation

across participating Selves. Human governance at population scope operates through joint authority across population-level governance perimeters, with the authority-versus-labor distinction from Paper 1 portable at this scope and the recursion bottoming at population-level human governance per Paper 1's governance-bottoms-at-human-authored-rules commitment.

Calibrated-humility register: Rung 6 is not a stated commitment of the same epistemic standing as Rungs 1–5. It is an extension claim. Population-scale CKS-governed Self deployments do not yet exist. Specific population-scale dynamics — convergence rates, differentiation patterns, governance bottlenecks at population scope — await empirical observation. What Rung 6 states is the architectural possibility: that the same substrate-mediator commitment which governs coordination at cell scope also governs the coordination surface that emerges at population scope when the ladder's inheritance is extended fully. The architectural shape of Rung 6 is rendered with the same structural confidence as Rungs 1–5. The empirical verification status of Rung 6 dynamics is rendered with calibrated humility that does not collapse into general hedging on the architectural commitments themselves.

Inheritance from Rungs 1–5: The inheritance structure at Rung 6 is the same as at every prior rung: all Paper 1 commitments carry forward with population-scope rendering. AI-as-substrate-mediator holds at population scope as LLMs operating as substrate mediators across many Selves; tool-agnosticism holds at the population substrate's structured-state-plus-access requirements; linear-cost composition holds at population-scale composition of coordination events rather than at within-cell composition of operations.

3. The ladder as prior-art structure

3.1 One architectural development, not six separate inventions

The prior-art significance of the ladder framework derives from a single structural claim: the six rungs are configurations of one architectural pattern at progressively larger scopes, not six separate architectural patterns composed into a sequence.

This distinction matters for how adversarial novelty claims land. If the six rungs were six separate inventions, a party could claim novelty for any individual rung by arguing that the others do not anticipate it. The ladder framework closes that argument. Rung 4 is what Rung 3 produces at enterprise scope; Rung 5 is what Rungs 3–4 produce at inter-Self scope; Rung 6 is what Rungs 1–5 produce at population scope. The architectural primitive — human-governed substrate as coordination medium — does not change across the rungs. The governance authority structure, the LLM's mediator role, the substrate's function as the coordination artifact, the conflict-preservation requirement, and the tool-agnosticism commitment all carry through every rung with only scope-level rendering differences.

The ladder is also observation rather than roadmap. Paper 1 did not promise Paper 3. Paper 2 explicitly bracketed inter-Self dynamics as out of scope. The ladder becomes visible in retrospect because Paper 3 completes it: articulating the sixth rung supplies the structural shape that makes the prior five visible

as a series. The series is not a forward prediction of where the architectural commitment was always heading; it is a backward reading of what the commitment composes to when extended at progressively larger scopes. This retrospective character is consistent with prior-art status — the architecture was there in the prior papers; the ladder formalization makes the structure explicit.

3.2 Parallel extension-claim structure across the trilogy

Each paper in the trilogy carries one core-theory rung and one extension-claim rung. Paper 1 contributes Rungs 1 and 2 (core theory and Claim 6 extension). Paper 2 contributes Rungs 3 and 4 (core theory and Claim 6 extension). Paper 3 contributes Rungs 5 and 6 (core theory and Claim 6 extension).

The parallel structure across the three Claim 6 positions — Paper 1’s Claim 6, Paper 2’s Claim 6, Paper 3’s Claim 6 — is what makes the trilogy’s extension claims architecturally honest. Each Claim 6 is the canonical extension claim at its paper’s scope: a projection of the prior rungs’ architecture onto the next larger coordination context, operating within the same substrate-mediator commitment, acknowledging calibrated humility at its own register. The parallelism is derived rather than imposed: the trilogy’s structure produces it.

3.3 Inheritance without discontinuity

A key structural property of the ladder is that governance architecture inherits from rung to rung without discontinuity. This means there is no rung at which the architectural primitive changes character, at which a new governance model replaces the prior one, or at which the LLM’s relationship to the substrate shifts from mediator to something else. The coordination scope enlarges at each rung; the governance architecture extends with it.

This property closes a class of adversarial claims: that inter-Self coordination (Rung 5) requires a governance model categorically different from cell-level governance (Rung 1), or that population-scale coordination (Rung 6) requires novel governance architectures not anticipated by the prior rungs. The inheritance chain from Rung 1 through Rung 6 means that each rung’s governance architecture is entailed by the prior rungs under the same architectural commitment.

4. Rungs 1–5 as commitments vs. Rung 6 in calibrated-humility register

The prior-art strength of the ladder differs between Rungs 1–5 and Rung 6, and this difference must be stated precisely.

Rungs 1–5 are directly defended by the core-theory claims of Papers 1, 2, and 3. Paper 1’s six core commitments (the source material for Rungs 1–2) receive detailed architectural defense across Papers 1 and 2 and are carried forward as Paper 3’s foundation. Paper 2’s core-theory claims (Rungs 3–4) receive full architectural defense within Paper 2. Paper 3’s Claims 1–5 (Rung 5) receive full architectural defense within Paper 3’s §§4–8. Adversarial challenges to any of Rungs 1–5 must engage

with the full architectural defense of the corresponding paper-level claims.

Rung 6 is the extension claim. Its prior-art status is real but its epistemic standing is different: the architectural *possibility* is stated, the inheritance from Rungs 1–5 is articulated, the three population-level commitments are named, and the calibrated-humility register is engaged because the empirical dynamics of population-scale CKS-governed Self coordination have not yet been observed. A challenge to Rung 6 can engage the architectural possibility framing; it cannot claim the population-scale architecture has no prior art, because the inheritance structure from Rungs 1–5 to Rung 6 is the prior art.

This distinction also matters for what the ladder formalization defends. The ladder as a prior-art structure defends the *relationship* among the rungs — the fact that they are one architectural development — at the full prior-art strength of Rungs 1–5. Rung 6’s position as the ladder’s completion is defended at the extension-claim register appropriate to Paper 3 Claim 6.

5. Three failure modes the sub-commitment defends against

Failure mode 1 — Treating the three papers as disconnected. If Papers 1, 2, and 3 are read as independent contributions, each paper’s claims can be evaluated in isolation from the others. The ladder formalization forecloses this reading: the three papers are one architectural development, and the prior-art force of each paper’s claims extends across the ladder to every rung the subsequent papers occupy. A novel-architecture claim for inter-Self coordination that ignores Paper 1’s Rung 1 commitments encounters those commitments through the ladder, not only through Paper 3.

Failure mode 2 — Claiming novelty for a higher rung while acknowledging lower rungs. An adversarial framing might concede Rung 1 (cell-level human-governed coordination) while claiming that inter-Self coordination (Rung 5) or enterprise integration (Rung 4) represents a novel architecture. The ladder closes this argument by establishing that each higher rung is an extension of the lower rungs under the same commitment, not a departure from them. The inheritance-without-discontinuity property (§3.3) is the structural basis for this closure.

Failure mode 3 — Population-scale AI coordination as requiring novel governance. A claim that population-scale AI coordination — across many organizations, many deployed AI systems, many governance perimeters — requires governance architectures not anticipated by prior work encounters Rung 6 as prior art for the architectural possibility that the same substrate-mediator governance applies at population scope. The calibrated-humility register does not weaken this prior-art status; it precisely characterizes what is claimed (architectural possibility, inheritance structure, population-scope rendering of Paper 1’s commitments) without overclaiming empirical dynamics that have not yet been observed.

6. Operational test

The ladder formalization produces the following operational test. For any deployed system that implements human–AI coordination at one or more of the six scopes, an observer can:

1. Identify which rungs are active in the deployment — which coordination scopes (cell, multi-human cell, intra-Self, enterprise Self, inter-Self, population) the system operates at.
2. For each active rung, verify that the governance architecture satisfies the corresponding paper’s architectural commitments: that the substrate functions as the coordination medium, that human authority (inspect, modify, override) is preserved at that scope, that the LLM acts as substrate mediator rather than as governance authority, that conflicts are preserved as first-class substrate state, that the architecture is tool-agnostic at the substrate layer, and that governance cost scales linearly with rule variety rather than with substrate size.
3. Trace the inheritance: verify that the governance architecture at each active rung is entailed by the governance architecture at the lower rungs active in the same deployment — that no rung introduces a discontinuity in the architectural primitive.

A deployment that satisfies this test across all its active rungs is implementing what the trilogy describes. A deployment that fails the test at any rung — because the substrate is not the coordination medium at that scope, because human authority is not preserved, because the LLM is the governance authority rather than the mediator, or because governance cost scales with substrate size rather than with rule variety — is not implementing the CKS architecture at that rung, regardless of how it describes itself.

For Rung 6, the test applies to the architectural possibility: a deployment that accumulates governed coordination events across many Selves at population scope and maintains the three rights at joint-authority scope across population-level governance perimeters satisfies Rung 6’s architectural commitment. The empirical dynamics of convergence, differentiation, and capability propagation at population scope are downstream of this architectural satisfaction, not constitutive of it.

7. Series T Phase T2 connection

This note formalizes the six-rung ladder as a framework — it establishes the ladder’s structure, the prior-art significance of each rung, the inheritance relationships among rungs, and the distinction between Rungs 1–5 and Rung 6. It does not formalize the transitions between rungs as standalone prior-art structures.

Each rung-to-rung transition — the architectural content that makes each rung an extension of the prior rung rather than a departure from it — will receive its own standalone prior-art note in Series T Phase T2 (notes T2.01 through T2.10). These notes will formalize the specific inheritance mechanisms, scope-level rendering differences, and governance-continuity properties that hold at each transition. Series T Phase T2 builds on the ladder framework this note establishes; D1.28 is the parent prior-art structure for the T2.xx transition notes.

8. Summary

The substrate-mediator ladder is the architectural spine of the CKS trilogy. Its six rungs, sources, and registers are as follows:

Rung	Scope	Source	Register
1	Single human ↔ LLM within one cell	Paper 1 core theory	Stated commitment
2	Multiple humans ↔ LLM within one cell	Paper 1 Claim 6	Stated commitment
3	Cell ↔ aspect ↔ Self within one Self	Paper 2 core theory	Stated commitment
4	Self as enterprise architecture	Paper 2 Claim 6	Stated commitment
5	Self ↔ Self via shared substrate	Paper 3 Claims 1–5	Stated commitment
6	Population-scale collective evolution	Paper 3 Claim 6	Calibrated-humility extension

Each rung is a configuration of one architectural primitive — human-governed substrate as coordination medium — at a progressively larger coordination scope. Each rung inherits the prior rungs’ governance architecture without discontinuity. The three papers are one architectural development. Rung 6 completes the ladder in the calibrated-humility register appropriate to an extension claim, without weakening the prior-art force of Rungs 1–5.

Derivation note #493 in the CKS defensive-publication series. Phase D1, note D1.28.